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Thermodynamic Modeling of the (Ce and Sm)-Ru Binary Systems Based on Linear, Exponential, and Combined Models Aided by Ab-Initio Calculations

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Abstract Thermodynamic optimization of the Ln-Ru (where Ln = Ce and Sm) binary systems was carried out by means of CALPHAD method. Our approach is based on the critical review of the available experimental data published in literature, including both phase equilibria and thermodynamic data. Nine intermediate phases:Ln_3Ru, Ce_7Ru_3, $Ce_{16}Ru_9$, Ce_4Ru_3, Sm_2Ru_5, $Ru_{25}Sm_{44}$, and $LnRu_2$ presented in both binary systems, were treated as stoichiometric compounds. The exponential and linear models were used to describe the temperature dependence of the excess quantities of liquid, (δCe), (γCe), (βCe), (αSm), (βSm), (γSm) and (Ru) solid solution phases. They were compared with the combined linear–exponential temperature dependence of the resulting interaction energies. The self-consistent thermodynamic parameters used to describe the Gibbs energies of various phases in the Ln-Ru binary systems were obtained through Thermo-Calc software. The obtained calculated results are in excellent agreement with available phase equilibrium and thermodynamic data.

Keywords CALPHAD · combined model · exponential model · first-principles calculations · linear model · ruthenium-rare earth alloys

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1 Introduction

Rare earth alloys with transition elements display notable properties which are the subject in development of several advanced materials, for instance material for hydrogen storage,[1,2] permanent magnets,[3] and cryogenic magnetic refrigeration.[4–10] Since Ce and Ru both have wide medical applications, the alloys of these two elements are of high demand in industry, for example, for manufacturing catalysts for oxidation in humid air[11] or other materials. Among the lanthanides, Ce plays a special role in that the otherwise localized *4f* orbital is partly itinerant in nature, which is caused by the relative shallowness of the *4f* orbital.[12,13] The low surface free energy improves surface sites for Ce, and cerium-transition metal catalyst systems are crucial for accelerating chemical reactions.[14,15]Ce is used as constituent in metal hydride catalysts,[16] namely, in the synthesis of ammonia, while, cerium oxide often serves as a catalyst in automotive exhaust purification.[17] To understand how cerium and samarium mix with ruthenium (the alloying behavior), an assessment was done with the aim of exploring potential uses for their magnetic properties. Consequently, the phase relations are the important benchmark for materials design and the vital basis for *CALPHAD* approach.[18–23] Our thermodynamic optimization of these two binary systems is

performed based on the experimental data available in the literature including phase diagram data, thermodynamic measurements, such as enthalpies of formation, and predicted data as well as derived ones from previous optimization made by Selhaoui et al.[24] using *Nancyun* program[25,26] based on polynomial functions to represent the properties of mixing of solution phases and a linear function of temperature to describe a stoichiometric compound. However, these approaches are too simplistic to be useful for binary systems. For complex systems necessary to use optimization programs which contain more advanced modeling, such as *Thermo-Calc* software developed by Jansson[27] and Sundman et al.[28]. Additionally, Chen et al.[29] noted the appearance of artificial inverted miscibility gaps in many calculated phase diagrams already published in literature. Theoretically, Kaptay[30] and Abe et al.[31] found that they appear if the linear model is used, and if $a_0^{\Phi} < 0$ and $b_0^{\Phi} > 2 \times R$. Kaptay's findings[32] are based on the Lupis-Elliott rule "Quite generally an increase in the temperature tends to bring a system closer to ideality", the statement is confirmed by showing that the heat and the excess entropy of mixing of a solution should have the same signs.[33] Kaptay, proposed a new function to describe the excess Gibbs free energy of the liquid and solution phases by assuming that all excess properties tend towards zero as temperature approaches infinity. Nevertheless, Schmid-Fetzer et al.[22] stated that the use of Kaptay equation to model the temperature dependence of the interaction parameters in the liquid and solid solution phases produces artificial phase stabilities at low temperatures. Then, Ref [30, 34] found that this problem can be avoided if the linear and exponential equations are coupled. The result complies with high-temperature boundary conditions.[32,33] To enhance the reliability of our assessments, we calculate the enthalpies of formation of each stoichiometric compound appearing in both systems using ab-initio. To summarize, the targeted objectives of our paper are to critically evaluate all the original experimental data and those derived from ab-initio calculations, and to compare the performance of the linear, the exponential and the combined models to describe the temperature dependence of the interaction energies in the RE-Ru(with RE = Ce and Sm) binary systems.

2 Literature Study on the Ln-Ru (with Ln = Ce and Sm) Phase Diagrams

2.1 Ce-Ru Phase Diagram

The first experimental determinations of phase equilibrium were made by Obrowski[35] and Reiswig and Gschneidner[36] based on thermal, micrographic and X-ray diffraction analyses. It was found that one solid compound ($CeRu_2$) with a $MgCu_2$-type structure and fcc parameter a = 7.5364 Å was formed. This compound melts peritectically at the temperature of 1570 °C (1843 K). A cerium-rich compound considered to be Ce_3Ru was reported by Palenzona[37]. A complete study was carried out by Palenzona[37] using several experimental methods such as: differential thermal analysis, X-ray diffraction analysis, electron microscopy and metallographic analysis. Thus, the existence of three binary compounds Ce_3Ru, Ce_7Ru_3, and $CeRu_2$ was confirmed. The Ce_2Ru and Ce_5Ru_3 binary compounds were reported with new stoichiometries, designated as $Ce_{16}Ru_9$ and Ce_4Ru_3, respectively. Later, the presence of Ce_4Ru was not checked. The solubility of ruthenium in cerium can be ignored since no noticeable effect on the transition temperature was observed. The solubility of cerium in ruthenium has not been studied because of the high melting point of the corresponding alloys, but according to Ref [38], it is less than 5%. As noted by Predel[39], the experimental procedure is well described and the experimental data resulting from several approaches are consistent with each other; the Ce-Ru phase diagram established by Palenzona[37] is the most reliable phase diagram present in the literature. It is depicted in Fig. 1(a). All intermediate phases Ce_3Ru, Ce_7Ru_3, $Ce_{16}Ru_9$, Ce_4Ru_3, and $CeRu_2$ are stoichiometric incongruent melting compounds at fixed compositions in addition to a succession of four peritectic reactions plateaus at 1843, 1023, 993, 1158, 868 K and one eutectic at 818 K falls on 0.15 at.% Ru. Subsequently, Okamoto[40] gave a corrected version of the phase diagram given by Palenzona[37], which was proposed in Massalski et al.[41] with a slight correction to the fact that maximal solubility of Ce in solid Ru at the peritectic temperature of $CeRu_2$ at 1843 K is about 2 at.%.

2.2 Sm-Ru Phase Diagram

The Sm-Ru phase diagram in Ref [41] was redrawn from Moffatt[42], which was based on intermediate phases reported in Ref [43–45], with a presumed similarity to *Gd*-Ru and *Dy*-Ru systems.[46] The entire liquidus boundaries were speculative. Palenzona and Canepa[47] determined the Sm-rich part of the phase diagram (0.6 to 1.0 at. Sm) with DTA measurements at thirteen compositions (see Fig. 1b). The "Sm_sRu_3" phase with a complex structure reported by Ref [45] was correct as $Sm_{44}Ru_{25}$, which is isomorphous to Y_{44} Ru_{25}. According to Ref [42], the allotropic transformation temperature of pure γSm to βSm is 1195 K whereas[47] found the transformation at 1218 K in alloys as well as in the pure element. Therefore, the reaction type of the (γSm) ↔ (βSm) transformation must be clarified. In addition, the liquidus line revised by Okamoto[48] is not a

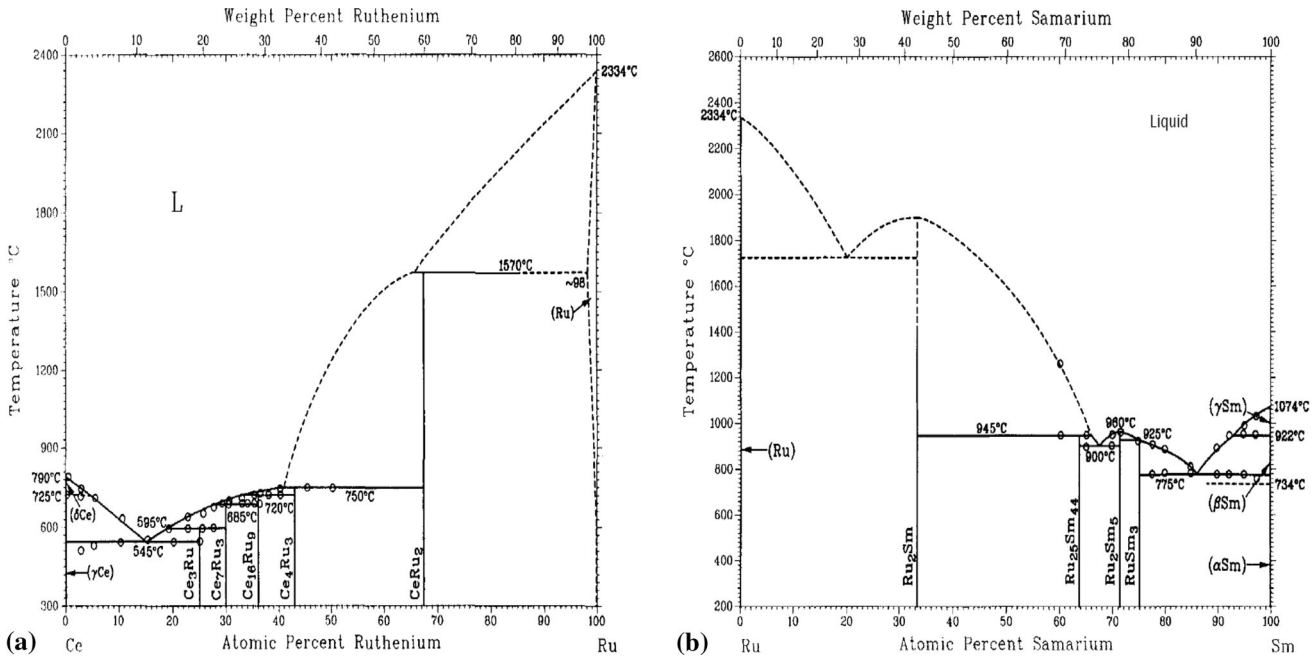

Fig. 1 (a) Experimental Cerium-Ruthenium phase diagram given by Palenzona[37] (o) and revised by Okamoto.[40] Published with permission Elsevier, Journal of Alloys and Compounds, 37, The Phase Diagram of the Ce-Ru System, Palenzona, A., 1991. (b) Experimental Samarium-Ruthenium phase diagram given by Palenzona and Canepa[47] (o) and revised by Okamoto.[48] Published with permission Elsevier, Journal of the Less-Common Metals, 47, Figure the Phase Diagram of the Sm-Ru System, Palenzona, A.; Canepa, F., 1989

solid but a dashed line, materializing the uncertainty of the experimental phase diagram. The crystal structures of compounds were originally determined by Ref [49] ($SmRu_2$, Cu_2Mg-type), Ref [47] (Sm_{44} Ru_{25}), Ref [44] (Sm_5 Ru_2), and Ref [43] (Sm_3Ru). Sm Ru_2 shows a $MgZn_2$-type high-pressure structure at > 6 kbar.[50] Crystal structure data and lattice parameter data for this system are given in Table 2.

All data concerning invariant reactions used for the assessment procedure for both systems are shown in Table 1. Information about crystal structure and lattice parameter data for the stable intermediate phases formed in the Ln-Ru (with Ln = Ce and Sm) binary systems are summarized in Table 2.

3 Thermodynamic Data

Due to the experimental intricacies such as high temperature and oxidation, the thermodynamic data of the Ln-Ru (with Ln = Ce and Sm) binary systems are scarce.

3.1 Ce-Ru System

There is only one published value of the standard enthalpy of formation of the $CeRu_2$ intermetallic compound measured by Selhaoui et Kleppa[51] using direct synthesis calorimetry at high temperature: ΔH_f ($CeRu_2$) = − 27.9 ± 1.5 kJ/mol-at. Furthermore, Selhaoui et al.[24] have estimated the melting entropies for the five stoichiometric compounds in order to have a sufficient number of equations. It should be mentioned that the reference state for Ce was modified as (βCe) for was used experimental value in Ref [51] and as (δCe) for calculation in Ref [24].

3.2 Sm-Ru System

There are no published measurements of the standard enthalpies of formation. In both cases, it is mandatory to estimate the enthalpies of formation of all compounds that exist in both systems. For this reason, we used the Miedema calculator,[52,53,69], which is a semi-empirical model for predicting thermochemical properties, especially for the enthalpies of formation of compounds and the enthalpies of mixing of solutions. However, the values given by Ref [52, 53, 69] are still significantly more exothermic than the previously measured values, as are our calculated results.

4 Ab-Initio Calculations

To enhance the available experimental data, we used first-principles calculations based on Density Functional Theory (DFT)[54,55] within the plane wave pseudopotentials as

Table 1 Special points in the Ce-Ru and Sm-Ru phase diagrams

Reaction	Composition, at.% Ru			Temperature, *K*	Reaction type	References
Ce-Ru binary system						
Liq ↔ (δCe)	0.00	0.00	…	1063	Melting	[37]
				1071		[41]
						This work^{LM}
						This work^{EM}
						This work^{CM}
						[24]
Liq + (δCe) ↔ (γCe)	4.80	0.00	0.00	999	Catatectic	[37]
	4.42			1000		This work^{LM}
	4.24			1000		This work^{EM}
	4.07			1000		This work^{CM}
	…			Does not appear		[24]
Liq ↔ (γCe) + Ce_3Ru	15.00	0.00	25.0	818.0	Eutectic	[37]
	13.85			820.5		This work^{LM}
	13.88			817.9		This work^{EM}
	14.35			818.3		This work^{CM}
	14.40			819.0		[24]
	14.30			814.0		
Liq + Ce_7Ru_3 ↔ Ce_3Ru	19.00	30.00	25.00	868.0	Peritectic	[37]
	17.56			864.5		This work^{LM}
	17.98			868.2		This work^{EM}
	18.53			868.1		This work^{CM}
	18.70			865.0		[24]
	18.60			865.0		
Liq + $Ce_{16}Ru_9$ ↔ Ce_7Ru_3	28.00	36.00	30.00	958.0	Peritectic	[37]
	28.82			961.1		This work^{LM}
	28.03			957.8		This work^{EM}
	28.46			957.0		This work^{CM}
	28.40			957.0		[24]
	27.50			950.0		
Liq + Ce_4Ru_3 ↔ $Ce_{16}Ru_9$	33.00	42.90	36.0	993.0	Peritectic	[37]
	34.46			989.9		This work^{LM}
	34.42			992.3		This work^{EM}
	34.79			992.1		This work^{CM}
	36.70			1007	$Ce_{16}Ru_{19}$ as Congruent	[24]
	36.30			1004		
Liq + $CeRu_2$ ↔ Ce_4Ru_3	41.00	66.70	42.90	1023	Peritectic	[37]
	40.55			1024		This work^{LM}
	40.24			1024		This work^{EM}
	40.36			1025		This work^{CM}
	41.00			1043		[24]
	41.00			1038		
Liq + (Ru) ↔ $CeRu_2$	66.00	98.00	66.70	1843	Peritectic	[37]
	65.84	98.17		1843		This work^{LM}
	67.04	96.79		1843		This work^{EM}
	66.10			1843		This work^{CM}
	66.10	100		1839		[24]
	66.00	100		1838		

Table 1 continued

Reaction	Composition, at.% Ru			Temperature, *K*	Reaction type	References
Sm-Ru binary system						
(αSm) ↔ (βSm)	0.00			1007	Allotropic	[47]
				1007		This work^{LM}
				1003		This work^{EM}
				1003		This work^{CM}
(βSm) ↔ (γSm)	0.00			1159		[47]
						This work^{LM}
						This work^{EM}
						This work^{CM}
Liq + (γSm) ↔ (βSm)	7.00	0.00	0.00	1195	Catatectic	[47]
						This work^{LM}
						This work^{EM}
						This work^{CM}
Liq ↔ (βSm) + Sm_3Ru	14.00	0.00	25.0	1048	Eutectic	[47]
	14.27			1048		This work^{LM}
	12.50			1050		This work^{EM}
	13.00			1048		This work^{CM}
Liq + Sm_5Ru_2 ↔ Sm_3Ru	25.00	28.50	25.00	1198	Peritectic	[47]
	24.21			1199		This work^{LM}
	22.88			1196		This work^{EM}
	23.46			1193		This work^{CM}
Liq ↔ Sm_5Ru_2	28.50	28.50	…	1233	Congruent	[47]
				1230		This work^{LM}
				1229		This work^{EM}
				1229		This work^{CM}
Liq ↔ Sm_5Ru_2 + $Sm_{44}Ru_{25}$	32.40	28.50	36.20	1173	Eutectic	[47]
	32.70			1199		This work^{LM}
	33.21			1207		This work^{EM}
	32.84			1204		This work^{CM}
Liq + $SmRu_2$ ↔ $Sm_{44}Ru_{25}$	34.00	66.70	36.20	1218	Peritectic	[47]
	34.32			1214		This work^{LM}
	34.00			1212		This work^{EM}
	34.30			1214		This work^{CM}
Liq ↔ $SmRu_2$	66.70	66.70	…	2174	Congruent	[47]
						This work^{LM}
						This work^{EM}
				2170		This work^{CM}
Liq ↔ $SmRu_2$ + (Ru)	79.80	66.70	100	2000	Eutectic	[47]
	81.80			2000		This work^{LM}
	79.72			2003		This work^{EM}
	79.90			2006		This work^{CM}
Liq ↔ (Ru)	100	100	…	2607	Melting	[37, 38, 47]
						This work^{LM}
						This work^{EM}
						This work^{CM}
						[24]

Table 2 Crystal structure and lattice parameters data of the stable intermediate binary phases in Ce-Ru and Sm-Ru binary systems

Phase /temperature range, °C	Composition, at.% Ru	Pearson symbol	Space group	Strukturbericht designation	Model type	Prototype	Phase model	Lattice parameters, nm			References
								a	*b*	*c*	
Ce-Ru binary system											
(δCe) 798-726	0.00	cI2	*Im* $\bar{3}m$	*A*2	SSM	W	$(Ce,Ru)_1$ $(Va)_2$	0.4120	…	…	[41]
(γCe) 726-61	0.00	cF4	*Fm* $\bar{3}m$	*A*1	SSM	Cu	$(Ce,Ru)_1$ $(Va)_1$	0.5161	…	…	
(βCe) 61-(-177)	0.00	hP4	*P6₃/mmc*	*A*3′	…	αLu	$(Ce,Ru)_1(Va)_{0.5}$*	0.3681	…	0.1185	
(αCe) < -177	0.00	cF4	*Fm* $\bar{3}m$	*A*1	…	Cu	$(Ce,Ru)_1(Va)_1$*	0.485	…	…	
Ce_3Ru < 595	25.0	oP16	*Pnma*	$D0_{11}$	SM	Fe_3C	$(Ce)_3$ (Ru)	0.7242	0.9863	0.6419	[37]
Ce_7Ru_3 < 685	30.0	hP20	*P6₃mc*	$D10_2$	SM	Th_7Fe_3	$(Ce)_7$ $(Ru)_3$	0.9802	…	0.6261	
								0.9864		0.6305	[42]
$Ce_{16}Ru_9$ < 720	36.0	hR150	*R* $\bar{3}m$	…	SM	…	$(Ce)_{16}$ $(Ru)_9$	1.3645[1]	…	2.2742	[37]
Ce_4Ru_3 < 750	42.9	mC28	*C2/m*	…	SM	…	$(Ce)_4$ $(Ru)_3$	0.8400	1.38370 β = 117.9°	0.5985	
$CeRu_2$ < 1570	66.7	cF24	*Fd* $\bar{3}m$	*C*15	SM	$MgCu_2$	$(Ce)_1$ $(Ru)_2$	0.7545	…	…	[35, 37, 42]
								0.7570	…	…	
								0.7779	…	…	
								0.7535	…	…	
Sm-Ru binary system											
(γSm) 922-1074	0.00	cI2	*Im* $\bar{3}m$	*A*2	SSM	W	$(Sm,Ru)_1$ $(Va)_1$	…	…	…	[41]
(βSm) 734-(922)	0.00	hP2	*P6₃/mmc*	*A*3	SSM	Mg	$(Sm,Ru)_1(Va)_{0.5}$*	0.3663	…	0.58448	
(αSm) < 734	0.00	hR3	*R* $\bar{3}m$	*C*19	SSM	αSm	…	0.3629	…	2.62071	
								0.3661	0.3661	0.8792	
								0.3628	0.3628	0.873	
Sm_3Ru < 925	25.0	oP16	*Pnma*	$D0_{11}$	SM	Fe_3C	$(Sm)_3$ (Ru)	0.7330	0.9461	0.6334	[44]
								0.7330	0.9508	0.6361	[47]

Table 2 continued

Phase /temperature range, °C	Composition, at.% Ru	Pearson symbol	Space group	Strukturbericht designation	Model type	Prototype	Phase model	Lattice parameters, nm			References
								a	*b*	*c*	
								0.7327	0.9500	0.6326	[45]
								0.7329	0.9477	0.6355	[47]
Sm_5Ru_2	28.6	mC28	*C2/c*	…	SM	…	$(Sm)_5$ $(Ru)_2$	1.6083	0.6483	0.7314	[44]
< 960									β = 96.89°		
								1.6111	0.6442	0.7295	[45]
									β = 96.87°		
								1.6045	0.6435	0.7298	[47]
									β = 96.84°		
$Sm_{44}Ru_{25}$	36.2	oP276	*Pnma*	…	SM	$Y_{44}Ru_{25}$	$(Sm)_{44}$ $(Ru)_{25}$	2.8339	1.5451	1.5451	[47]
< 945											
$SmRu_2$	66.7	cF24	$Fd\,\overline{3}m$	*C*15	SM	$MgCu_2$	$(Sm)_1$ $(Ru)_2$	0.7580	…	…	[49]
< 1901											
								0.7577	…	…	[50]
								0.7764	…	…	[45]
						$MgZn_2$		0.5298[2]	…	0.8939[2]	[50]
								0.5289[2]		0.8946[2]	[45]
(Ru)	100	hP2	$P6_3/mmc$	*A*3	SSM	Mg	$(Ln,Ru)_1$ $(Va)_3$	0.27058	…	0.42816	[41]
< 2334											
		cF4	$Fm\,\overline{3}m$	*A*1		Cu					

[1]Note a misprint in Ref [37] corrected here SSM: subregular solution model SM: sublattice model

[2]High pressure phase

implemented in Quantum Espresso code,[56] to determine the enthalpies of formation of the intermetallic compounds. The exchange correlation interaction was treated within the generalized gradient approximation (GGA), as given by Perdew, Burke and Ernzerhof (PBE).[57] A plane wave cutoff energy of 50 Ry is used for all considered compounds. The ab-initio calculations furnish the total energy at temperature = 0 K,E^{Φ}. The energy of formation is calculated by the following equation:

$$\Delta_f E = E^{\Phi} - \sum x_i \times E_i^0 \quad \text{(Eq 1)}$$

where E^0 is the total energy of element i in its stable state at 0 K. According to Colinet,[58] the energy of formation correspond to the enthalpy of formation.

The available experimental, interpolated, our ab-initio calculations data and those from online databases such as Materials Project,[59] OQMD,[60] Aflow,[61] and Jarvis,[62] enthalpies of formation of all solid compounds are tabulated in Table 3.

5 Thermodynamic Modeling

The Ln-Ru (with Ln = Ce and Sm) binary systems include the following phases: liquid, (δCe), (γCe), (βCe), (αSm), (βSm), (γSm) and (Ru) solid solution phases and nine stoichiometric compounds Ln_3Ru, Ce_7Ru_3, $Ce_{16}Ru_9$, Ce_4Ru_3, Sm_2Ru_5, $Ln_{25}Sm_{44}$, and $LnRu_2$.

5.1 Pure Elements

The pure elements in the two systems concerning liquid and seven solid solution phases, (δCe), (γCe), (βCe), (αSm), (βSm), (γSm) and (Ru), were described using the following equation:

$$G^{\Phi} - H_i^{\mathrm{SER}} = a + b \times T + c \times T \times \ln T + d \times T^2 + e \times T^3 + f \times T^{-1} + g \times T^7 + h \times T^{-9} \quad \text{(Eq 2)}$$

Where H_i^{SER} (298.15 K) is the molar enthalpy of the element i at 298.15 K and 10^5 Pa in its standard element reference (*SER*) state, fcc_A1 for Ce and Rhombohedral_C19 for (αSm) and T is the absolute temperature; *a, b, c, d, e, f, g* and *h* are coefficients. The last two terms in Eq 1 are only used for the metastable regions above and below the melting point, *g* for a liquid below the melting point and *h* for the solid phases above the melting point. The Gibbs free energy of the element i, in its SER state, is denoted by GHSER_i as follows:

$$\mathrm{GHSER_{Ru}} = {}^0G_{\mathrm{Ru}}^{\mathrm{hcp_A3}} = G_{\mathrm{Ru}}^{\mathrm{hcp_A3}} - H_{\mathrm{Ru}}^{\mathrm{hcp_A3}}\ (298.15\ \mathrm{K}) \quad \text{(Eq 3)}$$

$$\mathrm{GHSER_{Ce}} = {}^0G_{\mathrm{Ce}}^{\mathrm{fcc_A1}} = G_{\mathrm{Ce}}^{\mathrm{fcc_A1}} - H_{\mathrm{Ce}}^{\mathrm{fcc_A1}}\ (298.15\ \mathrm{K}) \quad \text{(Eq 4)}$$

$$\mathrm{GHSER_{Sm}} = {}^0G_{\mathrm{Sm}}^{\mathrm{Rhombohedral_C19}} = G_{\mathrm{Sm}}^{\mathrm{Rhombohedral_C19}} - H_{\mathrm{Sm}}^{\mathrm{Rhombohedral_C19}}\ (298.15\ \mathrm{K}) \quad \text{(Eq 5)}$$

In the present work, the Gibbs energies functions are taken from the *SGTE* (Scientific Group Thermodata Europe) database PURE (1, 3, and 4) from T = 298.15 to 10,000 K compiled by Dinsdale[63]

5.2 Solution Phases

The molar Gibbs energy of the solution phases can be expressed as the sum of the reference part, ideal part, excess part, and magnetic part:

$$G_m^{\Phi} = {}^{\mathrm{ref}}G^{\Phi} + {}^{\mathrm{id}}G^{\Phi} + {}^{\mathrm{ex}}G^{\Phi} + {}^{\mathrm{mag}}G^{\Phi} \quad \text{(Eq 6)}$$

$$G_m^{\Phi} = x_{\mathrm{Ln}}^{\Phi} \times {}^0G_{\mathrm{Ln}}^{\Phi} + x_{\mathrm{Ru}}^{\Phi} \times {}^0G_{\mathrm{Ru}}^{\Phi} + R \times T \times \left(x_{\mathrm{Ln}}^{\Phi} \times \ln x_{\mathrm{Ln}}^{\Phi} + x_{\mathrm{Ru}}^{\Phi} \times \ln x_{\mathrm{Ru}}^{\Phi}\right) + {}^{\mathrm{ex}}G^{\Phi} + {}^{\mathrm{mag}}G^{\Phi} \quad \text{(Eq 7)}$$

where R = 8.3145 J/mol-at.K is the universal gas constant, T (K) is the absolute temperature, x_{Ln} and x_{Ru} are the mole fraction of the elements Ln = Ce and Sm and Ru, respectively.

The excess Gibbs energy, ${}^{\mathrm{ex}}G_m^{\alpha}$ is described by using a Redlich–Kister type polynomial representation,[19,20,63,64] i.e.

$${}^{\mathrm{ex}}G^{\Phi} = x_{\mathrm{Ru}}^{\Phi} \times (1 - x_{\mathrm{Ru}}^{\Phi}) \times \sum_{j=0} L_j^{\Phi} \times (1 - 2 \times x_{\mathrm{Ru}}^{\Phi})^j \quad \text{(Eq 8)}$$

where the interaction energies of j-th order (usually j = 0,1,2) for the binary Ln-Ru solutions in phase Φ,L_j^{Φ} (J/mol-at.) are taken to vary linearly with temperature:

$$L_j^{\Phi} = a_j^{\Phi} + b_j^{\Phi} \times T \quad (j = 0, 1, 2, \ldots) \quad \text{(Eq 9)}$$

with the excess entropy $s = -b_j^{\Phi}$(J/mol-at.K) and the enthalpy $h = a_j^{\Phi}$ (J/mol-at.) for the j-th order interaction energy of the Ln-Ru solutions in phase Φ.

${}^{\mathrm{mag}}G^{\Phi}$ means the magnetic contribution to the molar Gibbs energy.[65] For both systems, during the optimization procedure, we did not take into account this contribution to the Gibbs energy of the intermetallic compounds and solution phases due to the following reasons: the Curie temperature of all the compounds in the Ln-Ru binary systems is far below 298.15 K, the compounds are in a

Table 3 The comparison between the optimized, calculated, predicted, interpolated, and measured enthalpies, entropies, and Gibbs free energies of formation of the ten solid compounds that exist in the Ce-Ru and Sm-Ru binary systems

Compounds	Ln	ΔH_f, kJ/mol-at.	ΔS_f, J/mol-at.K	ΔG_f at 1000 K, kJ/mol-at.	Method used	References
Ln_3Ru	Ce	− 19.34962	− 6.4063	− 12.94331	Optimization ^{LM}	Current work
		− 14.83605	− 4.50766	− 10.32838	Optimization ^{EM}	
		− 17.80096	− 4.97241	− 12.82854	Optimization ^{CM}	
		− 19.693	− 8.81	− 10.883	Calculation ^{I}	[24]
		− 25.143	− 16.58	− 8.563	Calculation ^{II}	
		− 23.53141	…	…	Miedema's model	[52, 53]
		− 20.0124	…	…	Ab-initio (Magnetic)	Current work
		− 20.00926	…	…	Ab-initio (No-magnetic)	
		− 19.28	…	…	Ab-initio	[59]
		− 12.9176	…	…	Ab-initio	[60]
		− 19.264	…	…	Ab-initio	[61]
		− 20.35269	…	…	Modified Miedema's model	[69]
	Sm	− 37.57609	− 1.9487	− 35.62739	Optimization ^{LM}	Current work
		− 21.30595	− 0.06146	− 21.24449	Optimization ^{EM}	
		− 21.47873	− 0.03663	− 21.44209	Optimization ^{CM}	
		− 26.59788	…	…	Miedema's model	[52, 53]
		− 23.24075	…	…	Modified Miedema's model	[69]
		− 23.031	…	…	Ab-initio	[59]
		− 22.261	…	…	Ab-initio	[60]
Ce_7Ru_3		− 23.1605	− 7.93641	− 15.22409	Optimization ^{LM}	Current work
		− 17.8141	− 5.75373	− 12.06037	Optimization ^{EM}	
		− 21.3837	− 6.41807	− 14.96563	Optimization ^{CM}	
		− 22.882	− 9.93	− 12.952	Calculation ^{I}	[24]
		− 28.425	− 18.08	− 10.345	Calculation ^{II}	
		− 28.30979	…	…	Miedema's model	[52, 53]
		− 26.3241	…	…	Ab-initio (Magnetic)	Current work
		− 26.41858	…	…	Ab-initio (No-magnetic)	
		− 26.22	…	…	Ab-initio	[59]
		− 12.4356	…	…	Ab-initio	[60]
		− 26.29536	…	…	Ab-initio	[61]
		− 23.30979	…	…	Modified Miedema's model	[69]
$Ce_{16}Ru_9$		− 27.86683	− 11.10514	− 16.76168	Optimization ^{LM}	Current work
		− 21.43206	− 8.16255	− 13.26951	Optimization ^{EM}	
		− 25.7403	− 9.29977	− 16.44052	Optimization ^{CM}	
		− 25.897	− 11.16	− 14.737	Calculation ^{I}	[24]
		− 31.341	− 19.44	− 11.901	Calculation ^{II}	
		− 33.86468	…	…	Miedema's model	[52, 53]
		− 29.329	…	…	Ab-initio	Current work
		− 31.908	…	…	Ab-initio	[59]
		− 7.9048	…	…	Ab-initio	[60]
		− 26.86102	…	…	Modified Miedema's model	[69]
Ce_4Ru_3		− 32.38663	− 14.58695	− 17.79968	Optimization ^{LM}	Current work
		− 25.62372	− 11.57311	− 14.05060	Optimization ^{EM}	
		− 30.78887	− 13.39515	− 17.39371	Optimization ^{CM}	
		− 28.082	− 12.13	− 15.952	Calculation ^{I}	[24]
		− 33.043	− 20.07	− 12.973	Calculation ^{II}	
		− 39.48561	…	…	Miedema's model	[52, 53]
		− 36.14638	…	…	Ab-initio (Magnetic)	Current work

Table 3 continued

Compounds	Ln	ΔH_f, kJ/mol-at.	ΔS_f, J/mol-at.K	ΔG_f at 1000 K, kJ/mol-at.	Method used	References
		− 36.0861	…	…	Ab-initio (No-magnetic)	
		− 36.053	…	…	Ab-initio	[59]
		− 8.4832	…	…	Ab-initio	[60]
		− 36.02368	…	…	Ab-initio	Current work
		− 29.36535	…	…	Modified Miedema's model	[69]
$LnRu_2$	Ce	− 18.90679	− 0.98435	− 17.92244	Optimization ^{LM}	Current work
		− 14.6208	− 0.29661	− 14.32418	Optimization ^{EM}	
		− 22.59881	− 4.50954	− 18.08927	Optimization ^{CM}	
		− 27.9 ± 1.5	…	…	Direct Synthesis Calorimetry	[51]
		− 29.72	− 11.34	− 18.38	Calculation ^{I}	[24]
		− 29.721	− 14.8	− 14.921	Calculation ^{II}	
		…	− 15	…	Estimation	
		− 41.97777	…	…	Miedema's model	[52, 53]
		− 27.19781	…	…	Ab-initio (Magnetic)	Current work
		− 27.19779	…	…	Ab-initio (No-magnetic)	
		− 27.281	…	…	Ab-initio	[59]
		− 30.848	…	…	Ab-initio	[60]
		− 28.0524	…	…	Ab-initio	[62]
		− 27.25856	…	…	Ab-initio	Current work
		− 32.22028	…	…	Modified Miedema's model	[69]
	Sm	− 66.9039	− 1.6456	− 65.2583	Optimization ^{LM}	Current work
		− 36.702	− 0.5529	− 36.1491	Optimization ^{EM}	
		− 34.01611	− 0.7669	− 33.24921	Optimization ^{CM}	
		− 45.79067	…	…	Miedema's model	[52, 53]
		− 36.03158	…	…	Modified Miedema's model	[69]
		− 19.177	…	…	Ab-initio	[59]
		− 21.297	…	…		
		− 17.828	…	…	Ab-initio	[61]
		− 20.044	…	…		
Sm_5Ru_2		− 39.9781	− 0.6238	− 39.35429	Optimization ^{LM}	Current work
		− 23.47266	− 0.01077	− 23.46188	Optimization ^{EM}	
		− 23.457	− 0.00069	− 23.4563	Optimization ^{CM}	
		− 30.4526	…		Miedema's model	[52, 53]
		− 25.9792	…		Modified Miedema's model	[69]
		− 25.922	…	…	Ab-initio	[59]
		− 25.055	…	…	Ab-initio	[60]
$Sm_{44}Ru_{25}$		− 45.30745	− 0.41058	− 44.89687	Optimization ^{LM}	Current work
		− 26.29098	− 0.11085	− 26.18013	Optimization ^{EM}	
		− 25.8035	− 0.16443	− 25.63907	Optimization ^{CM}	
		− 38.40541	…	…	Miedema's model	[52, 53]
		− 30.95779	…	…	Modified Miedema's model	[69]

paramagnetic state above this temperature and the thermodynamic functions of the Ln and Ru elements are given in a temperature range above 298.15 K.[63] Then, Kaptay[30,66] proposed a function to describe the temperature dependence of the excess Gibbs free energy of the solution and liquid phases. The binary interaction

parameter L_j^{Φ}, assessed in the present work, is temperature dependent, as follows:

$$L_j^{\Phi} = h_j^{\Phi} \times \exp\left(-\frac{T}{\tau_j^{\Phi}}\right) \quad \text{(Eq 10)}$$

where h_j^{Φ} (J/mol-at.) is the enthalpy part of the interaction energy extrapolated to $T = 0$ K, while τ_j^{Φ} (K) is a special temperature, at which the interaction energy would cross zero if it were described by the linear model (thus τ_j^{Φ} must always be positive).

By identifying h_j^{Φ} with a_j^{Φ} and τ_j^{Φ} with a_j^{Φ}/b_j^{Φ}, Eq 10 becomes Eq 11 with the excess entropy $b_j^{\Phi} = -s$:

$$L_j^{\Phi} = a_j^{\Phi} \times \exp\left(\left(-\frac{b_j^{\Phi}}{a_j^{\Phi}}\right) \times T\right) \quad \text{(Eq 11)}$$

According to Ref [30, 34], the linear and exponential equations are combined as:

$$L_j^{\Phi} = \left(h_j^{\Phi} - T \times s_j^{\Phi}\right) \times \exp\left(-\frac{T}{\tau_j^{\Phi}}\right) \quad \text{(Eq 12)}$$

where s_j^{Φ}(J/mol-at. K) is the entropy part of the interaction energy extrapolated to $T = 0$ K.

In the current work, we suggest comparing the performances of the linear, the exponential and the combined models to describe the temperature dependence of the interaction energies. We have limited the leading term of the interaction energy to $j = 0$. Thus, the excess Gibbs energy of the same solution Φ will be compared in three forms:

The linear model: $${}^{\text{ex}}G^{\Phi} = x \times (1-x) \times \left(h_0^{\Phi} - T \times s_0^{\Phi}\right) \quad \text{(Eq 13)}$$

The exponential model: $${}^{\text{ex}}G^{\Phi} = x \times (1-x) \times h_{e,0}^{\Phi} \times \exp\left(-\frac{T}{\tau_{e,0}^{\Phi}}\right) \quad \text{(Eq 14)}$$

The combined model: $${}^{\text{ex}}G^{\Phi} = x \times (1-x) \times (h_{c,0}^{\Phi} - T \times s_{0,c}^{\Phi}) \times \exp\left(-\frac{T}{\tau_{0,c}^{\Phi}}\right) \quad \text{(Eq 15)}$$

where h_0^{Φ} (J/mol-at.) and s_0^{Φ} (J/mol-at.K) are the temperature independent enthalpy and entropy terms of the linear model, $h_{e,0}^{\Phi}$ (J/mol-at.) and $\tau_{e,0}^{\Phi}$ (K) are the zero-Kelvin enthalpy term and the special temperature of the exponential model, while $h_{c,0}^{\Phi}$ (J/mol-at.), $s_{e,0}^{\Phi}$ (J/mol-at.K), and $\tau_{c,0}^{\Phi}$ (K) are the zero-Kelvin enthalpy terms, the zero-Kelvin entropy term and the special temperature of the combined model, respectively.

The artifact liquid solution re-stabilization at high temperatures takes place if thermodynamic condition $d^2\Delta G/dx^2 < 0$ is satisfied at $x_{\text{Ru}} = 0.5$. We derived the equations for the heat of mixing, excess entropy of mixing and excess heat capacity of mixing from the following identities:

$$S_{\Phi}^{E} = -\frac{dG_{\Phi}^{E}}{dT} \quad \text{(Eq 16)}$$

$$H_{\Phi} = G_{\Phi}^{E} - T \times \frac{dG_{\Phi}^{E}}{dT} \quad \text{(Eq 17)}$$

$$\Delta C_{p,\Phi}^{E} = -T \times \frac{d^2G_{\Phi}^{E}}{dT^2} \quad \text{(Eq 18)}$$

The excess term of the Gibbs energy was optimized for terminal solid solutions, due to solubility of Ln in Ru, which is reported in literature for Ce by Palenzona.[37] In the case of the three models, for fcc_A1 (Ce) and bcc_A2 (Ce) phases, ${}^0L$ was set equal to zero, i.e., a subregular solution approximation was adopted. To correctly describe a homogeneity range of hcp_A3 (Ru) phase, ${}^0L$ was used. In the case of the linear model, for the liquid phase, ${}^2L$ was necessary to reproduce the low-melting eutectic (at temperature of 1093 K) in the Ln-rich part of the Ln-Ru phase diagram.

5.3 Stoichiometric Compounds

Due to their negligible homogeneity ranges, all intermediate compounds Ln_3Ru, Ce_7Ru_3, $Ce_{16}Ru_9$, Ce_4Ru_3, Sm_5Ru_2, $Sm_{44}Ru_{25}$, $LnRu_2$, exist in Ln-Ru binary systems, and are treated in our optimization as stoichiometric phases, described as A_pB_q. The Gibbs energy of each phase is denoted as ${}^0G_{\text{Ln}_p\text{Ru}_q}$ and described by the following formula:

$${}^0G_{\text{Ln}_P\text{Ru}_q} = \frac{p}{p+q} \times {}^0G_{\text{Ln}} + \frac{q}{p+q} \times {}^0G_{\text{Ru}} + \Delta G_f^{\text{Ln}_p\text{Ru}_q} \quad \text{(Eq 19)}$$

where ${}^0G_{\text{Ln}}$ and ${}^0G_{\text{Ru}}$ is the Gibbs energy of the pure elements cerium, samarium, and ruthenium respectively; $\Delta G_f^{\text{Ln}_p\text{Ru}_q}$ is the Gibbs energy of formation per mole-atom of formula unit Ln_pRu_q. Due to a lack of experimental measurements, it is assumed that the Neumann–Kopp rule applies to the heat capacity, i.e. $\Delta C_P = 0$. Thus, $\Delta G_f^{\text{Ln}_p\text{Ru}_q}$ can be given by the following expression:

$$\Delta G_f^{\text{Ln}_p\text{Ru}_q} = a + b \times T \quad \text{(Eq 20)}$$

where the parameters a and b were to be optimized for each compound in the current work. The coefficient a can be estimated according to the corresponding experimental enthalpies of formation, and then evaluated by considering both phase equilibria and thermodynamic data (Table 4).

6 Optimization Procedure

In this section, we describe the adopted strategy for thermodynamic modeling of the studied systems. As a first step, we begin with the liquid phase, using the zeroth-order interaction parameters. Afterward, intermetallic compounds are inserted into the modeling one by one. We first introduced the thermodynamic parameters of the $CeRu_2$ compound based on enthalpy of formation data measured by Selhaoui and Kleppa,[51] using direct synthesis calorimetry at high temperature. During optimization, the first- and second-order interaction parameters of the liquid phase are introduced to describe the shape of liquidus in the phase diagrams. The calculated invariant temperatures are then adjusted to minimize the square sum of the errors between the experimental data and the computed values until most of the selected experimental information is reproduced within the expected uncertainty limits. The weights are adjusted systematically to ensure consistency of the data. The resulting thermodynamic parameters are presented in Table 5.

7 Results and Discussion

We treated the nine intermediate phases Ln_3Ru, Ce_7Ru_3, $Ce_{16}Ru_9$, Ce_4Ru_3, Sm_5Ru_2, $Sm_{44}Ru_{25}$, $LnRu_2$ in the Ln-Ru binary systems as stoichiometric compounds while the liquid phase and (Ru), (δCe), (γCe), (βCe), (αSm), (βSm) and (γSm) solid solutions were treated using a solution model. In both cases, we assessed the excess term of the Gibbs energy of the liquid and solution phases with (a) linear model,[18–23,64] (b) exponential model,[30,66] and (c) combined model.[34] A detailed analysis of our results is given below.

7.1 Ce-Ru Binary System

Figure 2(a) depicts the Ce-Ru phase diagram, calculated with the optimized thermodynamic parameters for the three models. It is very similar to the experimental one established by Palenzona[37] (see Fig. 1a). The calculated phase boundary of the liquid phase agrees well with the experimental data.[37] Also, the calculated liquidus in the Ce-rich part is in much better agreement with the experimental results than the one calculated in Ref [24]. A small difference is observed in the three cases concerning the liquidus in the composition range from 0.60 to 1.00 Ru. Table 1 shows the compositions and temperatures calculated for each reaction existing in the present system compared with the experimental data from Ref [37]. In the three cases, we have noted that all the invariant reactions are well reproduced, with an uncertainty about 2.4 K for temperature and 0.0186 at.% Ru for composition. It was necessary to mention that the Liq + $Ce_4Ru_3 \leftrightarrow Ce_{19}Ru_6$ peritectic reaction was calculated as the congruent melting point in both optimizations made by Selhaoui et al.[24]. The solubility of Ce in solid Ru at the Liq + hcp_A3 $\leftrightarrow$ $CeRu_2$ peritectic reaction at 1843 K is well reproduced with the three models: $\sim$ 1.83 at.% Ru for linear model and $\sim$ 3.21 at.% Ru for both exponential and combined models at 1843 K. Same remark goes for Liq + $CeRu_2 \leftrightarrow Ce_4Ru_3$ peritectic reaction which had calculated at (1024 K; 40.5 at.% Ru), (1024 K; 40.2 at.% Ru), and (1025 K; 40.3 at.% Ru) versus (1023 K; 41 at.% Ru) according to Ref [37] for the linear, exponential, and combined models, respectively. The two allotropic transitions (βCe) $\leftrightarrow$ (γCe) and (γCe) (δCe) are well reproduced at temperature of 334 and 998 K, respectively.

Table 4 The appearance temperature T_{min} of the artificial inverted miscibility gap due to the linear model for liquid phases in Ln-Ru (Ln = Ce and Sm) considered systems

Linear model[18–23, 64]						
Ln-Ru System	a_0^Φ, J/mol-at.		b_0^Φ, J/mol-at.K		$T_{min} = \frac{a_0^\Phi}{(2\times R - b_0^\Phi)}$, K	References
Ce-Ru	− 168,629.19	$a_0^\Phi < 0$	80.908	$b_0^\Phi > 2 \times R$	2623.350	[24]
	− 185,659.43		109.83		1992.000	
	− 108,981.27		27.600		9932.670	Current work
Sm-Ru	− 241,731.529		78.015		3937.893	

where R = 8.3145 J/mol K is the universal gas constant

Table 5 Thermodynamic parameters in the Ce-Ru and Sm-Ru binary systems[a]

Phase/temperature range, °C	Thermodynamic model	Used model**	Thermodynamic parameters[b]
Ce-Ru binary system			
Liquid	$(Ce,Ru)_1$	LM	${}^0L^{Liq}_{Ce,Ru} = -108981.27 + 27.59945 \times T$
			${}^1L^{Liq}_{Ce,Ru} = -4983.325 + 2.467602 \times T$
			${}^2L^{Liq}_{Ce,Ru} = 15081.73 + 2.880777 \times T$
		EM	${}^0L^{Liq}_{Ce,Ru} = -123889.49 \times \exp(-0.000597 \times T)$
			${}^1L^{Liq}_{Ce,Ru} = 48880.87 \times \exp(-0.001537 \times T)$
		CM	${}^0L^{Liq}_{Ce,Ru} = (-108840.7 + 27.44748 \times T) \times \exp(0.00000073 \times T)$
			${}^1L^{Liq}_{Ce,Ru} = (-45294.21 + 3.2337 \times T) \times \exp(-0.001963 \times T)$
			${}^2L^{Liq}_{Ce,Ru} = (33107.45 - 0.150678 \times T) \times \exp(-0.000403 \times T)$
(δCe) 798-726	$(Ce,Ru)_1(Va)*_3$	LM	No excess term optimized
		EM	
		CM	
(γCe) 726-61	$(Ce,Ru)_1(Va)*_1$	LM	
		EM	
		CM	
(βCe) 61-(-177)	$(Ce,Ru)_1(Va)_{0.5}{}^*$	LM	
		EM	
		CM	
(αCe) < -177	$(Ce,Ru)_1(Va)*_1$	LM	
		EM	
		CM	
(Ru) < 2334	$(Ce,Ru)_1(Va)_{0.5}{}^*$	LM	${}^0L^{HCP_A3}_{Ce,Ru} = -40918.93$
		EM	${}^0L^{HCP_A3}_{Ce,Ru} = -41473.58$
		CM	${}^0L^{HCP_A3}_{Ce,Ru} = -40918.93 \times \exp(0.00012 \times T)$
Ce_3Ru < 595	$Ce_{0.750}{:}Ru_{0.250}$	LM	${}^0G^{Ce_3Ru}_{Ce:Ru} - 0.750 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.750 \times {}^0G^{fcc_A1}_{Ce} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -19349.62 + 6.406 \times T$
		EM	${}^0G^{Ce_3Ru}_{Ce:Ru} - 0.750 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.750 \times {}^0G^{fcc_A1}_{Ce} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -14836.05 + 4.507 \times T$
		CM	${}^0G^{Ce_3Ru}_{Ce:Ru} - 0.750 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.750 \times {}^0G^{fcc_A1}_{Ce} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -17800.96 + 4.972 \times T$
Ce_7Ru_3 < 685	$Ce_{0.750}{:}Ru_{0.250}$	LM	${}^0G^{Ce_7Ru_3}_{Ce:Ru} - 0.700 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.300 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.700 \times {}^0G^{fcc_A1}_{Ce} + 0.300 \times {}^0G^{hcp_A3}_{Ru} = -23160.05 + 7.936 \times T$
		EM	${}^0G^{Ce_7Ru_3}_{Ce:Ru} - 0.700 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.300 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.700 \times {}^0G^{fcc_A1}_{Ce} + 0.300 \times {}^0G^{hcp_A3}_{Ru} = -17814.106 + 5.753 \times T$
		CM	${}^0G^{Ce_7Ru_3}_{Ce:Ru} - 0.700 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.300 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.700 \times {}^0G^{fcc_A1}_{Ce} + 0.300 \times {}^0G^{hcp_A3}_{Ru} = -21383.70 + 6.418 \times T$

Table 5 continued

Phase/temperature range, °C	Thermodynamic model	Used model**	Thermodynamic parameters[b]
$Ce_{16}Ru_9$ < 720	$Ce_{0.640}$:$Ru_{0.360}$	LM	${}^0G^{Ce_{16}Ru_9}_{Ce:Ru} - 0.640 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.360 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.640 \times {}^0G^{fcc_A1}_{Ce} + 0.360 \times {}^0G^{hcp_A3}_{Ru} = -27866.83 + 11.105 \times T$
		EM	${}^0G^{Ce_{16}Ru_9}_{Ce:Ru} - 0.640 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.360 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.640 \times {}^0G^{fcc_A1}_{Ce} + 0.360 \times {}^0G^{hcp_A3}_{Ru} = -21432.067 + 8.162 \times T$
		CM	${}^0G^{Ce_{16}Ru_9}_{Ce:Ru} - 0.640 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.360 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.640 \times {}^0G^{fcc_A1}_{Ce} + 0.360 \times {}^0G^{hcp_A3}_{Ru} = -25740.301 + 9.299 \times T$
Ce_4Ru_3 < 750	$Ce_{0.571}$:$Ru_{0.429}$	LM	${}^0G^{Ce_4Ru_3}_{Ce:Ru} - 0.571 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.429 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.571 \times {}^0G^{fcc_A1}_{Ce} + 0.429 \times {}^0G^{hcp_A3}_{Ru} = -32386.63 + 14.586 \times T$
		EM	${}^0G^{Ce_4Ru_3}_{Ce:Ru} - 0.571 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.429 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.571 \times {}^0G^{fcc_A1}_{Ce} + 0.429 \times {}^0G^{hcp_A3}_{Ru} = -25623.725 + 11.573 \times T$
		CM	${}^0G^{Ce_4Ru_3}_{Ce:Ru} - 0.571 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.429 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.571 \times {}^0G^{fcc_A1}_{Ce} + 0.429 \times {}^0G^{hcp_A3}_{Ru} = -30788.87 + 13.39 \times T$
$CeRu_2$ < 1570	$Ce_{0.333}$:$Ru_{0.667}$	LM	${}^0G^{CeRu_2}_{Ce:Ru} - 0.333 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.333 \times {}^0G^{fcc_A1}_{Ce} + 0.667 \times {}^0G^{hcp_A3}_{Ru} = -18906.799 + 0.984 \times T$
		EM	${}^0G^{CeRu_2}_{Ce:Ru} - 0.333 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.333 \times {}^0G^{fcc_A1}_{Ce} + 0.667 \times {}^0G^{hcp_A3}_{Ru} = -14620.8 + 0.2966 \times T$
		CM	${}^0G^{CeRu_2}_{Ce:Ru} - 0.333 \times {}^{298.15}H^{fcc_A1}_{Ce} - 0.667 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.333 \times {}^0G^{fcc_A1}_{Ce} + 0.667 \times {}^0G^{hcp_A3}_{Ru} = -22598.81 + 4.509 \times T$
Sm-Ru binary system			
Liquid	$(Sm,Ru)_1$	LM	${}^0L^{Liq}_{Sm,Ru} = -241731.5 + 78.015 \times T - 0.0343 \times T^2$ ${}^1L^{Liq}_{Sm,Ru} = 51841.92 - 39.886 \times T$ ${}^2L^{Liq}_{Sm,Ru} = 72735.02 - 10.893 \times T$
		EM	${}^0L^{Liq}_{Sm,Ru} = -123113.8 \times \exp(-0.000042 \times T)$ ${}^1L^{Liq}_{Sm,Ru} = -6171.209 \times \exp(0.000337 \times T)$ ${}^2L^{Liq}_{Sm,Ru} = 9614.009 \times \exp(0.000029 \times T)$
		CM	${}^0L^{Liq}_{Sm,Ru} = (-126311.8 + 0.03403 \times T) \times \exp(-0.000122 \times T)$ ${}^1L^{Liq}_{Sm,Ru} = (59519.06 - 36.763 \times T) \times \exp(-0.000153 \times T)$ ${}^2L^{Liq}_{Sm,Ru} = (2848546.9 + 4.6645 \times T) \times \exp(-0.00428 \times T)$
(βSm) > 1381	$(Sm,Ru)_1(Va)_3$*	LM	No excess term optimized
		EM	
		CM	
(αSm) 25-1381	$(Sm,Ru)_1(Va)_{0.5}$*	LM	
		EM	
		CM	
(Ru) < 2334	$(Sm,Ru)_1(Va)_{0.5}$*	LM	
		EM	
		CM	

Table 5 continued

Phase/temperature range, °C	Thermodynamic model	Used model**	Thermodynamic parameters[b]
$SmRu_3$ < 925	$Sm_{0.750}$:$Ru_{0.250}$	LM	${}^0G^{Sm_3Ru}_{Sm:Ru} - 0.750 \times {}^{298.15}H^{C19}_{Sm} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.750 \times {}^0G^{C19}_{Sm} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -37576.095 + 1.948 \times T$
		EM	${}^0G^{Sm_3Ru}_{Sm:Ru} - 0.750 \times {}^{298.15}H^{C19}_{Sm} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.750 \times {}^0G^{C19}_{Sm} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -36702.039 + 0.552 \times T$
		CM	${}^0G^{Sm_3Ru}_{Sm:Ru} - 0.750 \times {}^{298.15}H^{C19}_{Sm} - 0.250 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.750 \times {}^0G^{C19}_{Sm} + 0.250 \times {}^0G^{hcp_A3}_{Ru} = -21478.739 + 0.0366 \times T$
Sm_5Ru_2 < 960	$Sm_{0.714}$:$Ru_{0.286}$	LM	${}^0G^{Sm_5Ru_2}_{Sm:Ru} - 0.714 \times {}^{298.15}H^{C19}_{Sm} - 0.286 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.714 \times {}^0G^{C19}_{Sm} + 0.286 \times {}^0G^{hcp_A3}_{Ru} = -21305.951 + 0.061 \times T$
		EM	${}^0G^{Sm_5Ru_2}_{Sm:Ru} - 0.714 \times {}^{298.15}H^{C19}_{Sm} - 0.286 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.714 \times {}^0G^{C19}_{Sm} + 0.286 \times {}^0G^{hcp_A3}_{Ru} = -23472.66 + 0.0107 \times T$
		CM	${}^0G^{Sm_5Ru_2}_{Sm:Ru} - 0.714 \times {}^{298.15}H^{C19}_{Sm} - 0.286 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.714 \times {}^0G^{C19}_{Sm} + 0.286 \times {}^0G^{hcp_A3}_{Ru} = -23457.052 + 0.00069 \times T$
$Sm_{44}Ru_{25}$ < 945	$Sm_{0.638}$:$Ru_{0.362}$	LM	${}^0G^{Sm_{44}Ru_{25}}_{Sm:Ru} - 0.638 \times {}^{298.15}H^{C19}_{Sm} - 0.362 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.638 \times {}^0G^{C19}_{Sm} + 0.362 \times {}^0G^{hcp_A3}_{Ru} = -45307.45 + 0.4105 \times T$
		EM	${}^0G^{Sm_{44}Ru_{25}}_{Sm:Ru} - 0.638 \times {}^{298.15}H^{C19}_{Sm} - 0.362 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.638 \times {}^0G^{C19}_{Sm} + 0.362 \times {}^0G^{hcp_A3}_{Ru} = -26290.98 + 0.11 \times T$
		CM	${}^0G^{Sm_{44}Ru_{25}}_{Sm:Ru} - 0.638 \times {}^{298.15}H^{C19}_{Sm} - 0.362 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.638 \times {}^0G^{C19}_{Sm} + 0.362 \times {}^0G^{hcp_A3}_{Ru} = -25803.507 + 0.164 \times T$
$SmRu_2$ 1700-1900	$Sm_{0.333}$:$Ru_{0.667}$	LM	${}^0G^{SmRu_2}_{Sm:Ru} - 0.667 \times {}^{298.15}H^{C19}_{Sm} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.667 \times {}^0G^{C19}_{Sm} + 0.333 \times {}^0G^{hcp_A3}_{Ru} = -66903.964 + 1.645 \times T$
		EM	${}^0G^{SmRu_2}_{Sm:Ru} - 0.667 \times {}^{298.15}H^{C19}_{Sm} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.667 \times {}^0G^{C19}_{Sm} + 0.333 \times {}^0G^{hcp_A3}_{Ru} = -36702.039 + 0.552 \times T$
		CM	${}^0G^{SmRu_2}_{Sm:Ru} - 0.667 \times {}^{298.15}H^{C19}_{Sm} - 0.333 \times {}^{298.15}H^{hcp_A3}_{Ru} = +0.667 \times {}^0G^{C19}_{Sm} + 0.333 \times {}^0G^{hcp_A3}_{Ru} = -34016.117 + 0.766 \times T$

Units in J/mol.at. and J/mol-at.K. Reference states are solid (γCe) for Ce, (hcp_A3) for Ru, and (αSm) for Sm at T[a]

[a]Since the Neumann–Kopp Rule was used the formation properties are independent of temperature

[b]Note that the enthalpies and entropies of formation of the intermetallic compounds are listed per mole of atoms

*(V_a) for Vacancy

**LM, linear model; EM, exponential model; CM, combined model

7.2 Sm-Ru Binary System

We calculated the Sm-Ru phase diagram using three models based on optimized thermodynamic parameters (see Fig. 2b). All invariant equilibria are calculated without a significant discrepancy in temperatures or compositions. The phase transformations (αSm) ↔ (βSm) and (βSm) (γSm), and peritectic reaction Liq + (γSm) ↔ (βSm) are well reproduced with the three models. The latter is calculated at 1195 K. The greatest discrepancy is obtained for the eutectic reaction Liq ↔ Sm_5Ru_2 + $Sm_{44}Ru_{25}$ for which the following values are obtained: 1199 K and 32.7 at.% Rufrom the linear model; 1207 K and 33.21 at.% Rufrom the exponential model and 1204 K and 32.84 at.% Rufrom the combined model. According to Ref [47, 48], these values are 1173 K and 32.4 at.% Ru.

7.3 Comparison of our Thermodynamic Results and the Literature Data

We optimized all thermodynamic parameters together to give the best description. Table 3 compares, the literature

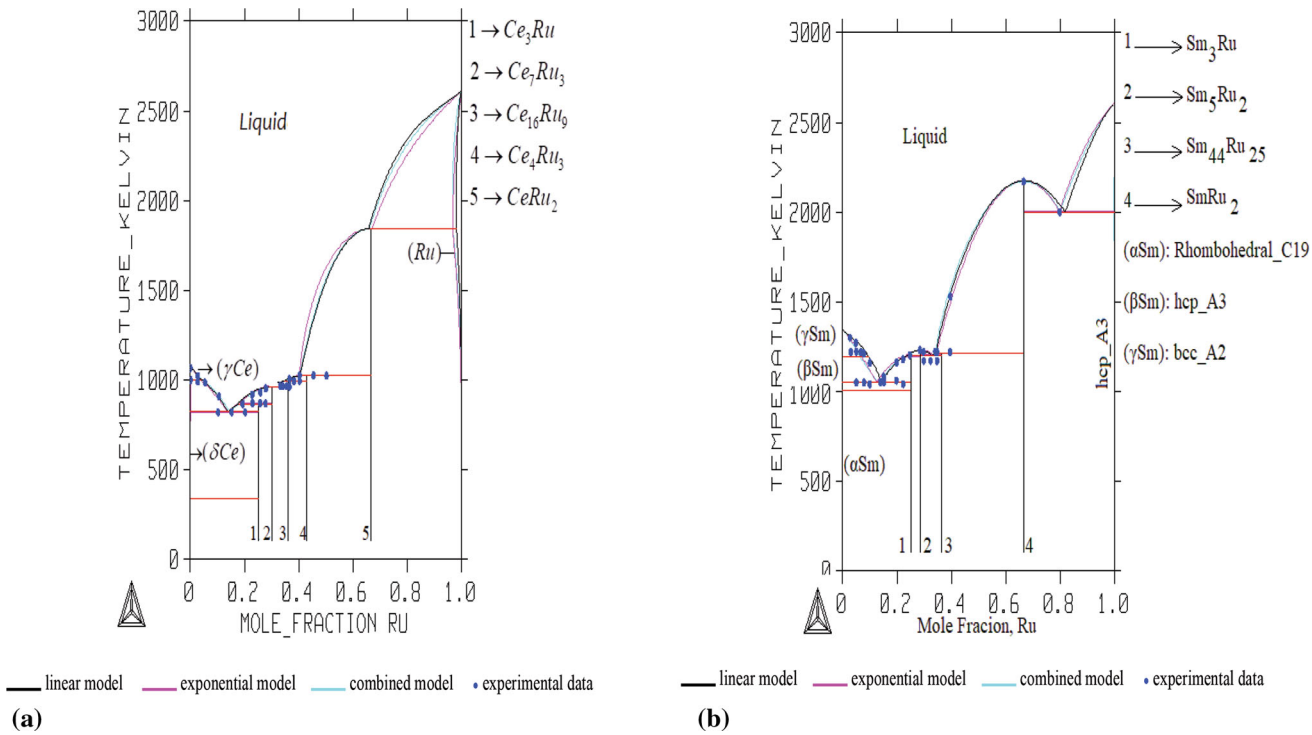

Fig. 2 (a). Superimposed Ce-Ru binary phase diagram calculated using our thermodynamic parameters for the three cases: linear model, exponential model, and combined model, compared to experimental data established by Palenzona.[37] Published with permission Elsevier, Journal of Alloys and Compounds, 37, The Phase Diagram of the Ce-Ru System, Palenzona, A., 1991. (b). Superimposed Sm-Ru binary phase diagram calculated using our thermodynamic parameters for the three cases: linear model, exponential model, and combined model, compared to experimental data from Ref [47] and revised version by Okamoto.[48] Published with permission El Sevier, Journal of the Less-Common Metals, 47, Figure the Phase Diagram of the Sm-Ru System, Palenzona, A.; Canepa, F., 1989

enthalpies of formation, measured by the calorimetric methods,[51,67,68,70–72] predicted by Ref [52, 53, 69], and calculated by ab-initio method from Materials project,[59] OQMD,[60] Aflow[61] and Jarvis,[62] with our optimized enthalpies of formation of the intermetallic compounds in the Ce-Ru and Sm-Ru binary systems. Figure 3(a) and (b) display the enthalpies, Gibbs energies, and relative Gibbs energies of formation of the intermetallic compounds existing in Ln-Ru systems referred to solid Ln and solid Ru, that are optimized in this work by CALPHAD method with the three models. Table 3 compares these thermodynamic values with the previously known ones. They are less negative than those predicted in Ref [52, 53, 69]. The enthalpy of formation of the $LnRu_2$ is the highest. For that phase, the calculated enthalpy is in complete agreement with the experimental data determined.[51,67,68,70–72] All values are given in kJ/mol-at. In the case of Ce-Ru system, it must be noted that the $CeRu_2$ compound is the only one for which the enthalpy of formation was measured by direct synthesis calorimetry.[51] Our enthalpy of formation of this compound is slightly more negative than the experimental one. The predicted standard enthalpies of formation of five compounds in this system were calculated using the Miedema model.[52,53,69] They are also shown. This model gives large uncertainties. The enthalpies of formation obtained from our ab-initio calculations are more exothermic than our calculated values from the three models. The discrepancies with the entropies of formation in literature are higher compared to our calculated values and ones obtained from both calculations in Ref [24]. In the previous study, these values are more negative and deviate from the general rule $-10 \leq \Delta S_f$ (J/mol − at.K) ≤ 10.

As shown in Fig. 3(b), our calculated enthalpy of formation for the $SmRu_2$ phase with the combined model are more exothermic than those calculated with linear and exponential models, and values derived from literature.

In this section, it is appropriate to evaluate the thermodynamic stability of the Gibbs energy of $Ln_{1-x}Ru_x$ intermediate phases with respect to $Ln_{1-x'}Ru_{x'}$ and $Ln_{1-x''}Ru_{x''}$ neighboring phases. Such a comparison can be made based on the formula given by Voronin:[73]

$$\Delta_r G(x) = G(x) - (1-\eta) \times G(x') - \eta \times G(x'') \quad \text{(Eq 21)}$$

where $x'' \geq x \geq x'$ are the mole fractions of Ru in the corresponding phases and $\eta = {(x - x')}/{(x'' - x')}$.

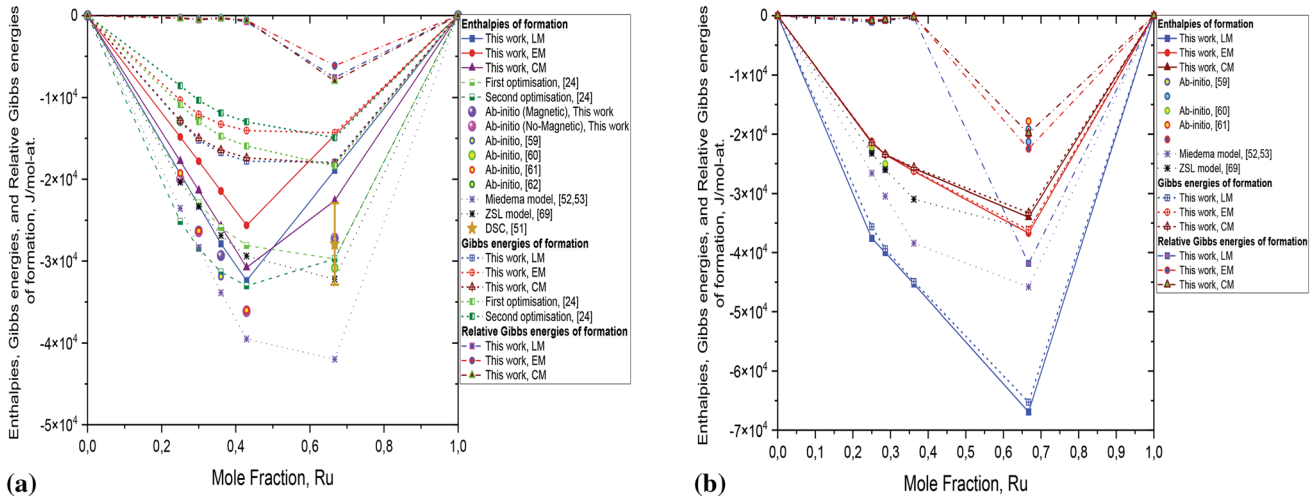

Fig. 3 (a) Relative Gibbs energies, Gibbs energies and standard enthalpies of formation for alloys of Ce with Ru compared with data for CeRu2 by Selhaoui and Kleppa,[51] our ab-initio calculations for the five intermetallic compounds together with ab-initio data from Ref [59–62], predicted values from the original model of Miedema and co-workers,[52,53], and ZSL model by Zhang.[69] (b). Relative Gibbs energies, Gibbs energies and standard enthalpies of formation for alloys of Sm with Ru compared with ab-initio data from Ref [59–62], predicted values from the original model of Miedema and co-workers,[52,53] and ZSL model by Zhang[69]

The calculation results are shown in Fig. 3(a) and (b). As follows from the calculation, the $LnRu_2$ phase is the most thermodynamically stable. According to the Landolt-Börnstein reference book,[39] all Ln-Ru phase diagrams, except for the *Eu*-Ru system, form the $LnRu_2$ phase. The heavy rare earth elements from *Gd* to *Lu* form a congruently melting phase $LnRu_2$ of the $MgCu_2$ type with a cubic cell $Fd\overline{3}m$. All these phases melt approximately at temperatures of 2070 ± 50 K.

The phases based on $LnRu_2$ from La to Sm have a hexagonal unit cell $P6_3/mmc$ of the $MgZn_2$ type. The nature of the melting of $LnRu_2$ phases of the $MgCu_2$ type remains open. According to Ref [39], the $LaRu_2$, $CeRu_2$, and $PrRu_2$ are formed by peritectic reaction in the temperature range 1400–1431, 1570–1573 K and 1954 K, respectively. According to Ref [38], the $NdRu_2$ phase has a peritectic melting point at 1635 K, while according to Ref [41], it melts congruently at 2173 K. The $SmRu_2$ phase undoubtedly melts congruently at 2173 K.

Based on the assessment of the thermodynamic stability of the $LnRu_2$ phases relative to the phases coexisting with it, according to formula (21), the phases in the series $LaRu_2$, $CeRu_2$, $PrRu_2$, and $SmRu_2$ are the most thermodynamically stable, as shown in Fig. 3(a) and (b), and must have congruent melting temperatures, which increase almost linearly. At the same time, the Gibbs energies also change almost linearly depending on their melting points. However, since the congruent temperatures of the $LnRu_2$ phases are unknown, and the coordinates of the $LnRu_2$-Ru eutectic are also unknown, we took these phases as incongruent. Such an assumption is possible if the congruent temperatures are close to the temperatures of peritectic reactions, and the eutectic are located near the $LnRu_2$ phases as in the case when Ln = Nd.

Table 5 lists the final optimized thermodynamic parameters for both systems, using the three models. As we can see, in agreement with Zhu et al.,[74] the enthalpy to formation for different Laves phases can vary significantly, indicating the different bonding features and electronic configurations in the Laves phases.

Moreover, we have plotted the excess Gibbs energy of mixing for the liquid phase at $x_{Ru} = 0.5$ as a function of the temperature from 0 to 20,000 K for L_0^{Φ} by applying the linear, exponential, and combined models as follows (for the Ce-Ru system):

Redlich Kistler model:

$$\Delta G^E_{x_{Ru}} = x_{Ru} \times (1 - x_{Ru}) \times (-1.089812E + 05 + 27.599 \times T) \quad \text{(Eq 22a)}$$

$$\Delta H = x_{Ru} \times (1 - x_{Ru}) \times (-1.089812E + 05) \quad \text{(Eq 22b)}$$

$$\Delta S^E_{x_{Ru}} = x_{Ru} \times (1 - x_{Ru}) \times (-27.59945) \quad \text{(Eq 22c)}$$

$$\Delta C^E_{p,x_{Ru}} = 0 \quad \text{(Eq 22d)}$$

Kaptay model:

$$\Delta G^E_{x_{Ru}} = x_{Ru} \times (1 - x_{Ru}) \times (-1.238894E + 05 \times \mathrm{EXP}(-5.976979E - 04 \times T)) \quad \text{(Eq 23a)}$$

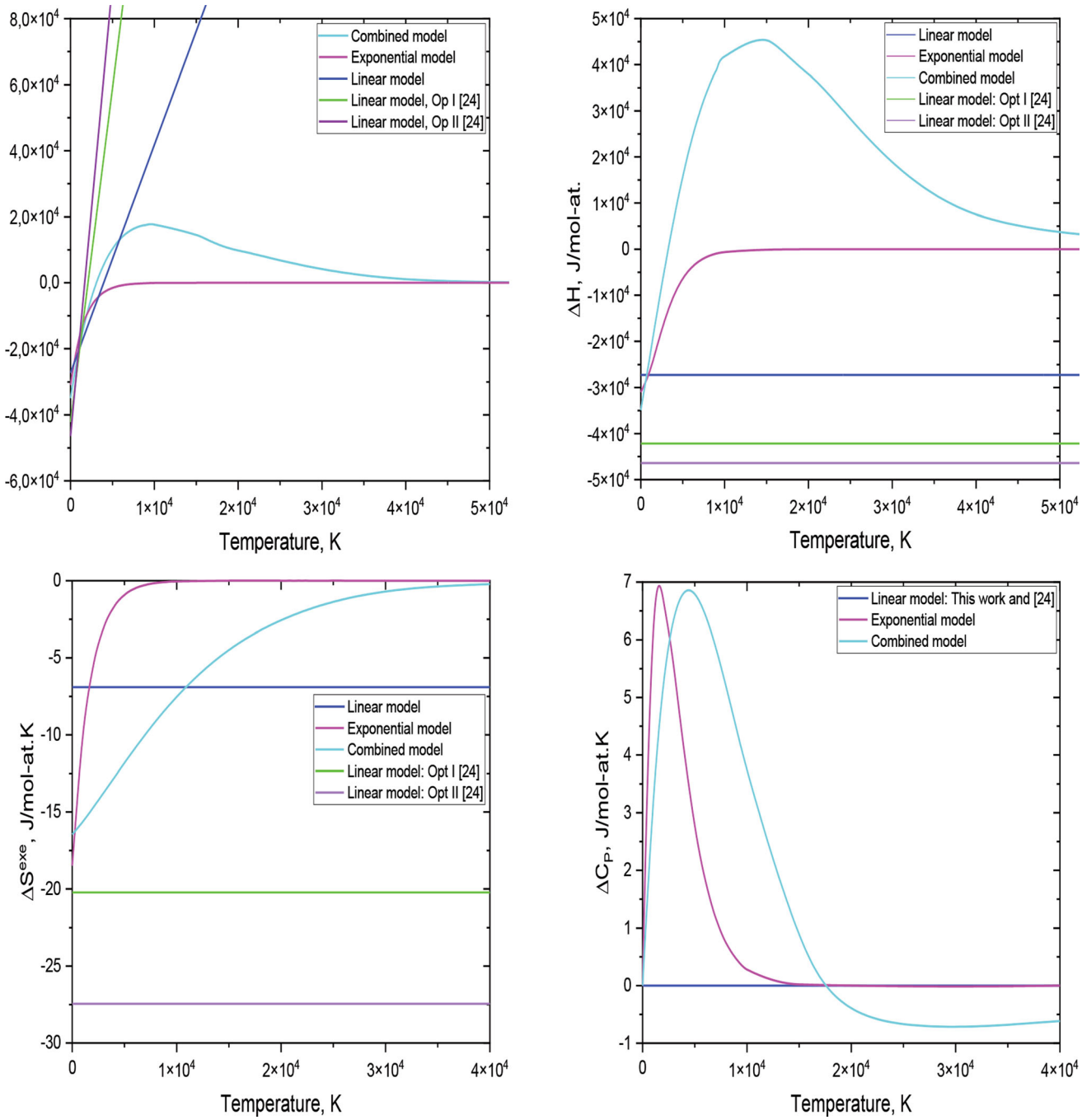

Fig. 4 Calculated T-dependencies of the excess functions for an equiatomic solution for Cerium-Ruthenium system using the linear model (Eq 9, 22a–c), and from Ref [24] for both optimization I and II), of the exponential model (Eq 10, 23a–c), and of the combined model (Eq 12, 24a–c)

$$\Delta H = x_{\mathrm{Ru}} \times (1 - x_{\mathrm{Ru}}) \times (-1.238894E + 05 \times (1 + 5.976979E - 04 \times T) \times \mathrm{EXP}(-5.976979E - 04 \times T)) \quad \text{(Eq 23b)}$$

$$\Delta S^{E}_{x_{\mathrm{Ru}}} = -x_{\mathrm{Ru}} \times (1 - x_{\mathrm{Ru}}) \times (73.982) \times \mathrm{EXP}(-5.976979E - 04 \times T) \quad \text{(Eq 23c)}$$

$$\Delta C^{E}_{p,x_{\mathrm{Ru}}} = x_{\mathrm{Ru}} \times (1 - x_{\mathrm{Ru}}) \times (44.123E - 03) \times T \times \mathrm{EXP}(-5.976979E - 04 \times T) \quad \text{(Eq 23d)}$$

Combined model:

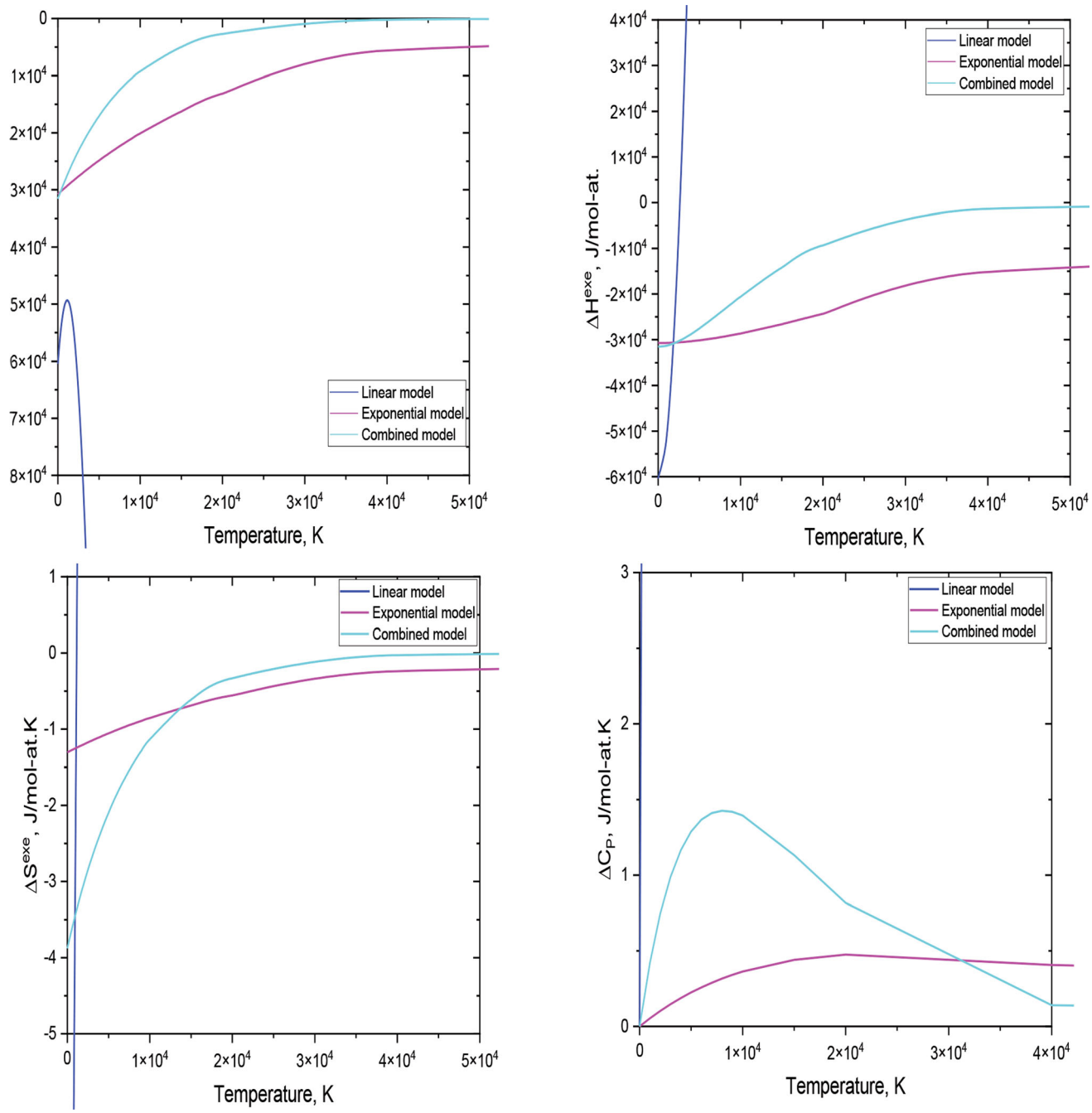

Fig. 5 Calculated T-dependencies of the excess functions for an equiatomic solution for Samarium-Ruthenium system using the linear model (Eq 9), of the exponential model (Eq 10), and of the combined model (Eq 12)

$$\Delta G^{E}_{x_{\mathrm{Ru}}} = x_{\mathrm{Ru}} \times (1 - x_{\mathrm{Ru}}) \times (-1.397264E + 05 + 45.128 \times T) \times \mathrm{EXP}(-1.484E - 04 \times T) \quad \text{(Eq 24a)}$$

$$\Delta H = x_{\mathrm{Ru}} \times (1 - x_{\mathrm{Ru}}) \times (-1.39726E + 05) \times (1 + 1.484E - 04 \times T - 4.79E - 08 \times T^{2}) \times \mathrm{EXP}(-1.484E - 04 \times T) \quad \text{(Eq 24b)}$$

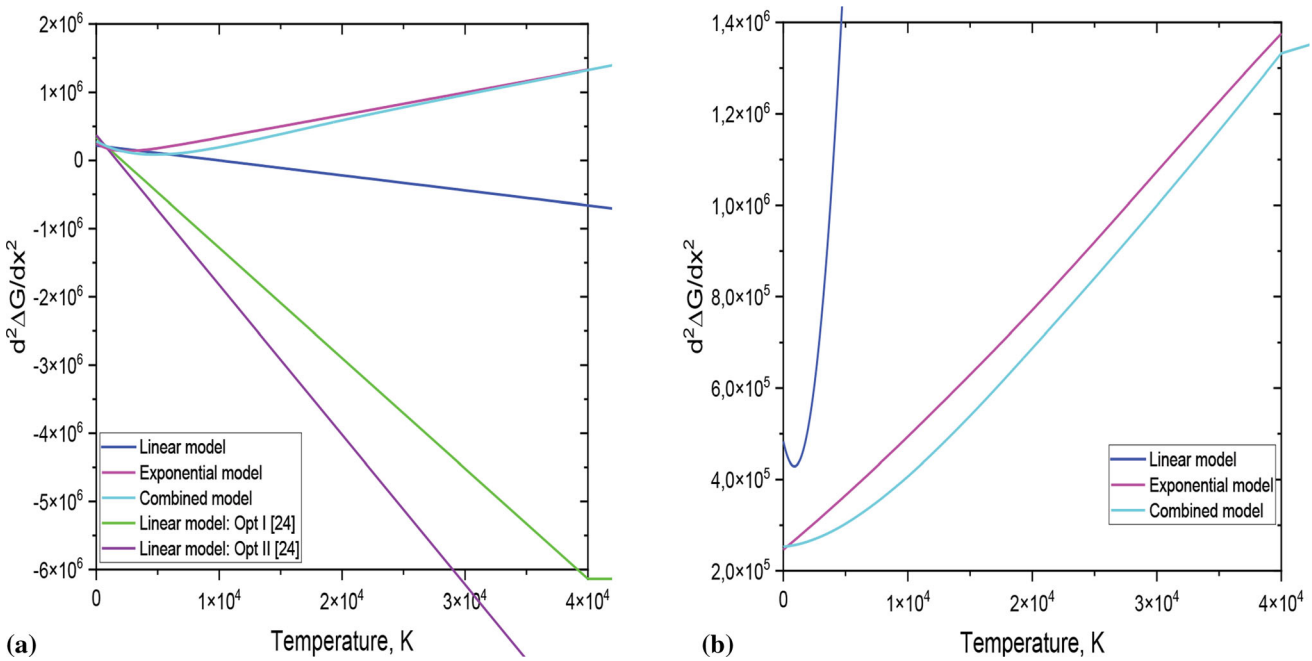

Fig. 6 Calculated second derivative vs. temperature for an equiatomic solution for Cerium-Ruthenium (a) and Samarium-Ruthenium (b) systems using the exponential model, combined model, and linear model

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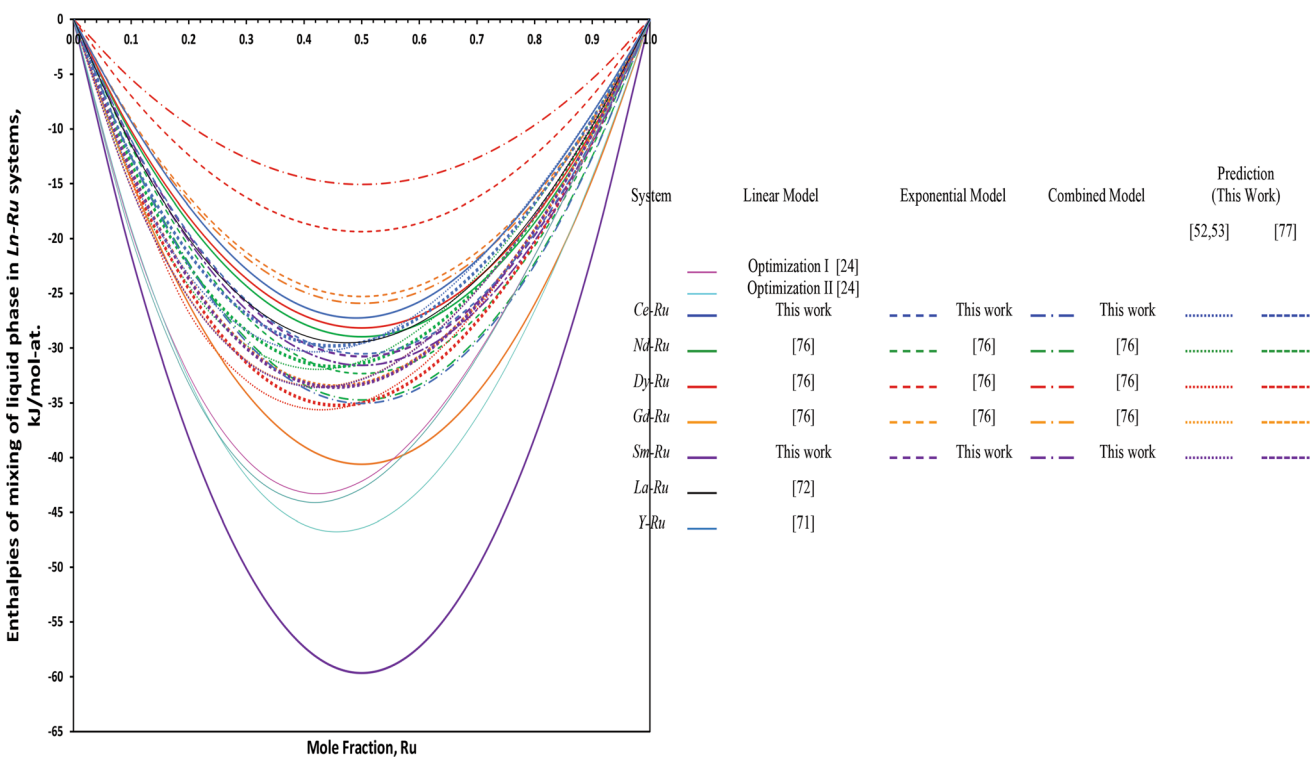

Fig. 7 Calculated liquid enthalpies of mixing at $T = 1000$ K in the Ln-Ru binary systems where Ln = Ce,[37] Sm,[47,48] Y,[71] La,[72] Nd,[38] Dy,[46] Gd,[46] and Sm[47,48] compared with those issued from first and second optimizations by Ref [24], from Ref [76], predicted by Miedema model,[52,53] and approximated by Takeuchi and Inoue.[77] The references states are: liquid for both Ln and Ru

$$\Delta S^{E}_{x_{\mathrm{Ru}}} = -x_{\mathrm{Ru}} \times (1 - x_{\mathrm{Ru}}) \times (45.128) \times (1.459 - 1.484E - 04 \times T) \times \mathrm{EXP}(-1.484E - 04 \times T) \quad \text{(Eq 24c)}$$

$$\Delta C^{E}_{p,x_{\mathrm{Ru}}} = -x_{\mathrm{Ru}} \times (1 - x_{\mathrm{Ru}}) \times (66.97E - 04 \times T) \times (-2.459 + 1.48E - 04 \times T) \times \mathrm{EXP}(-1.48E - 04 \times T) \quad \text{(Eq 24d)}$$

Figure 4 and 5, demonstrate that in the three cases for both systems, the four excess properties are continuous functions of temperature in the whole temperature interval at any composition. At high temperatures for exponential and combined models the excess Gibbs energy function tends toward zero in accordance with the rule of Lupis and Elliott[33] and later reformulated in Ref [32]. This behavior is in agreement with the requirements given in Ref [66]. However, for the linear model, this function changes its sign from negative to positive and goes to more and more positive values, which is probably nonphysical.[32] Furthermore, the linear model predicts $\Delta C^{E}_{p,x_{\mathrm{Ru}}} = 0$, which is in contradiction with the principal trend of the excess heat capacity of mixing as mentioned in Ref [34]. At a medium temperature range, all excess functions cross zero and go through maxima. At a low temperature, the linear model provides more negative excess Gibbs energy value compared to the exponential and combined models. This behavior changes at intermediate temperatures; the two latter models provide more negative excess Gibbs energy compared to the first one. The behavior observed with the linear model can be explained by using the temperature (T^2) factor d of Eq 1. As can be seen the ΔG^E and ΔC_p excess properties tend towards negatives values at intermediate temperatures. At the same time, the ΔH function tends toward positives values when temperature increases. The same behavior was observed for ΔS^E function. Moreover, we have plotted the $\mathrm{d}^2\Delta G/\mathrm{d}x^2$ at $x_{\mathrm{Ru}} = 0.5$ as a function of temperature from 0 to 15,000 K for both systems for the three models. The results are shown in Fig. 6(a) and (b). With the linear model, for the Ce-Ru system, we detect the occurrence of miscibility gap at high temperature ($a_0^{\Phi} < 0$ and $b_0^{\Phi} > 2 \times R$ are satisfied); this arises from $T_{\min}$ (see Table 4), while for the other systems, we observe $\mathrm{d}^2\Delta G/\mathrm{d}x^2 > 0$ in the entire considered temperature range. This is due to the optimized term (d) related to T^2 in the excess Gibbs energy of the liquid phase. The solution will not be stable at $T > a_0^{\Phi}/(2 \times R - b_0^{\Phi})$. This finding is in accordance with Kaptay.[34] Due to the application of an oversimplified linear model,[18–23,64] a considerable amount of systems suffer from a high temperature miscibility gap. To systematically analyze the mixing enthalpy of liquid Ln-Ru (where Ln = Ce, Nd, Sm, Gd, and *Dy*) alloys, we give the mixing enthalpies of liquid Y-Ru and La-Ru alloys measured and calculated by Selhaoui et al.[71,72,75] together with our recent calculated results from Ref [76] in Fig. 7. The references states are liquid for all elements at 1873 K under a pressure of 10^5 Pa. With decreasing RE atomic numbers, the calculated enthalpies of mixing of liquid in Ln-Ru (Ln = Ce, Nd, Sm, Gd, and *Dy*) alloys become much more negative. Besides, Takeuchi and Inoue[77] approximated the enthalpy of mixing with a formula of a subregular solution model with an interaction parameter Ω due to the awkward procedures for evaluating surface fraction term in Miedema's scheme. We use this approximation to calculate the enthalpy of mixing of liquid phase in our system (where Ω_k with k = 0,1,2, and 3 values were taken from Ref [77]) together with one predicted by Miedema model.[52,53] The calculated enthalpies of mixing of liquid Ce-Ru alloys have a negative deviation and show minimum values in the composition range of 40–50 at.% Ru. To check the occurrence of a low temperature artifact due to the exponential model in this system, we provide the mixing Gibbs energy of the liquid and Gibbs energy of formation of the five stoichiometric compounds at T = 298.15 K in Fig. 8. Functions are referred to the structures hcp_A3 (ruthenium) and bcc_A2 (cerium). As we can see, the exponential model for Gibbs energy of

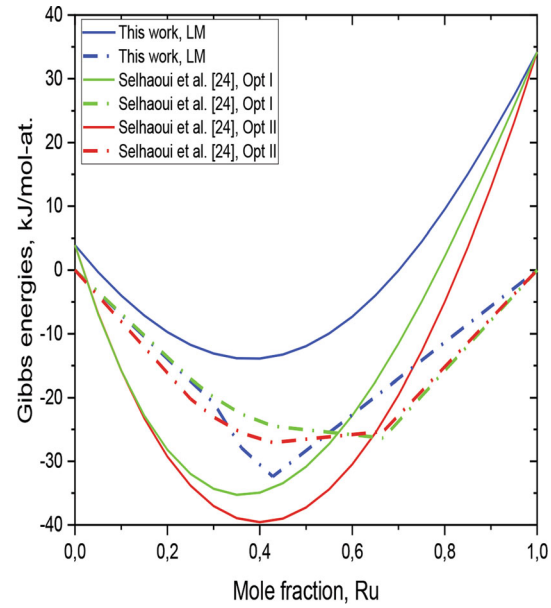

Fig. 8 Calculated Gibbs free energies of mixing of liquid phase and of the stochiometric compounds exist in Cerium-Ruthenium system using the linear model (blue solid and dash dot lines) compared with those calculated by Selhaoui et al.:[24] Opt I (green solid and dash dot lines) and Opt II (red solid and dash dot lines) at temperature of 298.15 K. References state are bcc_A2 for Ce and hcp_A3 for Ru

solid phases provides more negative values compared to the Gibbs energy of the liquid phase.

8 Conclusion

We demonstrate the usefulness of the combined linear-exponential model to avoid CALPHAD artifacts at any temperature. Despite the limited amount of experimental data such as enthalpies of mixing of the liquid phases, self-consistent thermodynamic descriptions were obtained. We used them to calculate their corresponding phase diagrams, which are generally in reasonable agreement with the data available in the literature.

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